SECTION 1: Identification

1.1. GHS Product identifier

Product form	:	Substance
Trade name	:	ETİFERT B-20
Chemical name	:	Disodium octaborate tetrahydrate
IUPAC name	:	Disodium octaborate
Substance type	:	Mono-constituent
EC-No.	:	234-541-0
EC Index-No.	:	005-020-00-3
CAS-No.	:	12280-03-04
Formula	:	Na2B8O13.4H2O
Product group	:	Trade product

1.2. Other means of identification

No additional information available

1.3. Recommended use of the chemical and restrictions on use

Recommended uses and restrictions	:	Not restricted
Recommended use	:	Agriculture (Fertilizer, Formulation of fertilizer) Flame-retardant agent Wood articles

1.4. Supplier's details

Manufacturer

ETİ MADEN İŞLETMELERİ GENEL MÜDÜRLÜĞÜ
Kızılırmak Mahallesi 1443. Cadde No:5 06530
Çukurambar-Çankaya
Ankara - TÜRKİYE
T +90 312 294 20 00 - F +90 312 230 71 84
info@etimaden.gov.tr - www.etimaden.gov.tr

1.5. Emergency phone number

+90 312 294 20 00

SECTION 2: Hazard identification

2.1. Classification of the substance or mixture

Classification according to the United Nations GHS

Acute toxicity (oral), Category 5	H303
Acute toxicity (dermal), Category 5	H313
Reproductive toxicity, Category 2	H361d
Hazardous to the aquatic environment – Acute Hazard, Category 3	H402

Full text of H statements : see section 16

Adverse physicochemical, human health and environmental effects	:	May be harmful if swallowed or in contact with skin, Suspected of damaging the unborn child, Harmful to aquatic life

2.2. GHS Label elements, including precautionary statements

Labelling according to the United Nations GHS

<tx:_GHS_PICTOS> (GHS UN) :



Signal word (GHS UN)	:	Warning
Hazard statements (GHS UN)	:	H303+H313 - May be harmful if swallowed or in contact with skin H361d - Suspected of damaging the unborn child H402 - Harmful to aquatic life

Precautionary statements (GHS UN) : P201 - Obtain special instructions before use.
P202 - Do not handle until all safety precautions have been read and understood.
P273 - Avoid release to the environment.
P280 - Wear protective gloves/protective clothing/eye protection/face protection/hearing protection/....
P308+P313 - IF exposed or concerned: Get medical advice/attention.
P312 - Call a POISON CENTER/doctor/physician if you feel unwell.
P405 - Store locked up.

2.3. Other hazards which do not result in classification

Other hazards not contributing to the classification : Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

SECTION 3: Composition/information on ingredients

3.1. Substances

Substance type : Mono-constituent
IUPAC name : Disodium octaborate
Chemical name : Disodium octaborate tetrahydrate
Product identifiers: See section 1.1

Name	Product identifier	%	Classification according to the United Nations GHS
Disodium octaborate tetrahydrate (Main constituent)	(CAS-No.) 12280-03-04	> 99.25	Acute Tox. 5 (Oral), H303 Acute Tox. 5 (Dermal), H313 Repr. 2, H361d Aquatic Acute 3, H402

Full text of H-statements: see section 16

3.2. Mixtures

Not applicable

SECTION 4: First-aid measures

4.1. Description of necessary first-aid measures

First-aid measures general : IF exposed or concerned: Get medical advice/attention. Call a poison center or a doctor if you feel unwell.

First-aid measures after inhalation : Remove person to fresh air and keep comfortable for breathing.

First-aid measures after skin contact : Wash skin with plenty of water. Take off contaminated clothing.

First-aid measures after eye contact : Rinse eyes with water as a precaution.

First-aid measures after ingestion : Rinse mouth. Call a poison center or a doctor if you feel unwell.

4.2. Most important symptoms/effects, acute and delayed

No additional information available

4.3. Indication of immediate medical attention and special treatment needed, if necessary

Treat symptomatically.

SECTION 5: Fire-fighting measures

5.1. Suitable extinguishing media

Suitable extinguishing media : Water spray. Dry powder. Foam.

5.2. Specific hazards arising from the chemical

Hazardous decomposition products in case of fire : Toxic fumes may be released.

5.3. Special protective actions for fire-fighters

Protection during firefighting : Do not attempt to take action without suitable protective equipment. Self-contained breathing apparatus. Complete protective clothing.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

6.1.1. For non-emergency personnel

Emergency procedures : Ventilate spillage area. Avoid contact with skin, eyes and clothing.

6.1.2. For emergency responders

Protective equipment : Do not attempt to take action without suitable protective equipment. For further information refer to section 8: "Exposure controls/personal protection".

6.2. Environmental precautions

Avoid release to the environment.

6.3. Methods and materials for containment and cleaning up

For containment : Collect spillage.

Methods for cleaning up : Mechanically recover the product. Notify authorities if product enters sewers or public waters.

Other information : Dispose of materials or solid residues at an authorized site.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Precautions for safe handling : Ensure good ventilation of the work station. Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear personal protective equipment. Do not get in eyes, on skin, or on clothing.

Hygiene measures : Wash contaminated clothing before reuse. Do not eat, drink or smoke when using this product. Always wash hands after handling the product.

7.2. Conditions for safe storage, including any incompatibilities

Storage conditions : Store locked up. Store in a well-ventilated place. Keep cool.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

ETİFERT B-20 (12280-03-04)	
DNEL/DMEL (Workers)	
Long-term - systemic effects, dermal	316.4 mg/kg bodyweight/day
Long-term - systemic effects, inhalation	6.7 mg/m³
DNEL/DMEL (General population)	
Acute - systemic effects, oral	0.79 mg/kg bodyweight/day
Long-term - systemic effects,oral	0.79 mg/kg bodyweight/day
Long-term - systemic effects, inhalation	3.4 mg/m³
Long-term - systemic effects, dermal	159.5 mg/kg bodyweight/day
PNEC (Water)	
PNEC aqua (freshwater)	2.9 mg/l
PNEC aqua (marine water)	2.9 mg/l
PNEC aqua (intermittent, freshwater)	13.7 mg/l
PNEC (Soil)	
PNEC soil	5.7 mg/kg dwt
PNEC (STP)	
PNEC sewage treatment plant	10 mg/l

8.2. Appropriate engineering controls

Appropriate engineering controls : Ensure good ventilation of the work station.

Environmental exposure controls : Avoid release to the environment.

8.3. Individual protection measures, such as personal protective equipment (PPE)

Hand protection : Protective gloves

Eye protection : Safety glasses

Skin and body protection : Wear suitable protective clothing

Respiratory protection : [In case of inadequate ventilation] wear respiratory protection.

Personal protective equipment symbol(s)



8.4. Exposure limit values for the other components

No additional information available

SECTION 9: Physical and chemical properties

9.1. Basic physical and chemical properties

Physical state : Solid

Appearance : White solid

Molecular mass	:	412.5 g/mol
Colour	:	Not available
Odour	:	odourless.
Odour threshold	:	Not available
Melting point	:	815 °C
Freezing point	:	Not applicable
Boiling point	:	Not available
Flammability (solid, gas)	:	Non flammable.
Explosive limits	:	Not applicable
Lower explosive limit (LEL)	:	Not applicable
Upper explosive limit (UEL)	:	Not applicable
Flash point	:	Not applicable
Auto-ignition temperature	:	Not applicable
Decomposition temperature	:	No data available
pH	:	8.5 @25 °C
pH solution	:	1 %
Viscosity, kinematic (calculated value) (40 °C)	:	Not applicable
Partition coefficient n-octanol/water (Log Kow)	:	Not available
Vapour pressure	:	Negligible @20 °C
Vapour pressure at 50 °C	:	Not available
Density	:	Not available
Relative density	:	Not available
Relative vapour density at 20 °C	:	Not applicable
Solubility	:	Water: 17.65g/100g @20 °C; 27.08g/100g @40 °C
Particle size	:	Not available
Particle size distribution	:	Not available
Particle shape	:	Not available
Particle aspect ratio	:	Not available
Particle specific surface area	:	Not available

9.2. Data relevant with regard to physical hazard classes (supplemental)
No additional information available

SECTION 10: Stability and reactivity
10.1. Reactivity
The product is non-reactive under normal conditions of use, storage and transport.

10.2. Chemical stability
Stable under normal conditions.

10.3. Possibility of hazardous reactions
No dangerous reactions known under normal conditions of use.

10.4. Conditions to avoid
None under recommended storage and handling conditions (see section 7).

10.5. Incompatible materials
No additional information available

10.6. Hazardous decomposition products
Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information
11.1. Information on toxicological effects
Acute toxicity (oral)	:	May be harmful if swallowed.
Acute toxicity (dermal)	:	May be harmful in contact with skin.
Acute toxicity (inhalation)	:	Not classified

ETİFERT B-20 (12280-03-04)	
LD50 dermal rabbit	> 2000 mg/kg bodyweight Animal: rabbit, Guideline: other:FIFRA (40 CFR 163)

Skin corrosion/irritation : Not classified
pH: 8.5 @25 °C

Serious eye damage/irritation : Not classified
pH: 8.5 @25 °C

Respiratory or skin sensitisation : Not classified

Germ cell mutagenicity : Not classified

Carcinogenicity : Not classified

Reproductive toxicity : Suspected of damaging the unborn child.

STOT-single exposure : Not classified

STOT-repeated exposure : Not classified

Aspiration hazard : Not classified

ETİFERT B-20 (12280-03-04)	
Viscosity, kinematic	Not applicable

SECTION 12: Ecological information

12.1. Toxicity

Ecology - general : Harmful to aquatic life.

Hazardous to the aquatic environment, short–term (acute) : Harmful to aquatic life.

Hazardous to the aquatic environment, long–term (chronic) : Not classified

ETİFERT B-20 (12280-03-04)	
LC50 - Fish [1]	74 mg/l Test organisms (species): Limanda limanda
LC50 - Fish [2]	79.7 mg/l Test organisms (species): Pimephales promelas
EC50 72h - Algae [1]	66 mg/l Test organisms (species): Phaeodactylum tricornutum
EC50 72h - Algae [2]	54 mg/l Test organisms (species): Phaeodactylum tricornutum
NOEC chronic fish	6.4 mg/l Test organisms (species): Danio rerio (previous name: Brachydanio rerio) Duration: '34 d'

12.2. Persistence and degradability

ETİFERT B-20 (12280-03-04)	
Persistence and degradability	Boron is naturally occurring and ubiquitous in the environment. Disodium octaborate tetrahydrate decomposes in the environment to natural borate.

12.3. Bioaccumulative potential

ETİFERT B-20 (12280-03-04)	
Bioaccumulative potential	Not bioaccumulative.

12.4. Mobility in soil

ETİFERT B-20 (12280-03-04)	
Mobility in soil	This product is soluble in water and is leachable through normal soil.

12.5. Other adverse effects

Ozone : Not classified

Other adverse effects : No additional information available

SECTION 13: Disposal considerations

13.1. Disposal methods

Waste treatment methods : Dispose of contents/container in accordance with licensed collector's sorting instructions.

SECTION 14: Transport information

In accordance with IMDG / IATA / UN RTDG

UN RTDG	IMDG	IATA
14.1. UN number		
Not regulated for transport		
14.2. UN Proper Shipping Name		
Not applicable	Not applicable	Not applicable
14.3. Transport hazard class(es)		
Not applicable	Not applicable	Not applicable
Not applicable	Not applicable	Not applicable
14.4. Packing group		
Not applicable	Not applicable	Not applicable
14.5. Environmental hazards		
Dangerous for the environment : No	Dangerous for the environment : No Marine pollutant : No	Dangerous for the environment : No
No supplementary information available		

14.6. Special precautions for user

- UN RTDG
No data available
- IMDG
No data available
- IATA
No data available

14.7. Transport in bulk according to IMO instruments

Not applicable

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations specific for the product in question

Regulatory reference : Listed on IECSC (Inventory of Existing Chemical Substances Produced or Imported in China)
Listed on KECL/KECI (Korean Existing Chemicals Inventory)
Listed introduction on Australian Industrial Chemicals Introduction Scheme (AICIS Inventory)
Listed on PICCS (Philippines Inventory of Chemicals and Chemical Substances)
Listed on NZIoC (New Zealand Inventory of Chemicals)
Listed on the Canadian DSL (Domestic Substances List)
Listed on the EEC inventory EINECS (European Inventory of Existing Commercial Chemical Substances)
Listed on the Japanese ENCS (Existing New Chemical Substances) inventory
Listed on the United States TSCA (Toxic Substances Control Act) inventory.

SECTION 16: Other information

Issue date : January 5, 2017
Revision date : January 20, 2023

Section	Changed item	Change	Comments
	Precautionary statements (GHS UN)	Modified	P273 - Avoid release to the environment.
	Hazard statements (GHS UN)	Modified	H303+H313 - May be harmful if swallowed or in contact with skin

			H402 - Harmful to aquatic life
2.1	Adverse physicochemical, human health and environmental effects	Modified	May be harmful if swallowed or in contact with skin, Harmful to aquatic life
2.1	Classification (GHS UN)	Modified	Acute toxicity (dermal), Category 5 H313 Hazardous to the aquatic environment – Acute Hazard, Category 3 H402

Abbreviations and acronyms : CAS-No. - Chemical Abstract Service number
ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE - Acute Toxicity Estimate
BCF - Bioconcentration factor
BLV - Biological limit value
BOD - Biochemical oxygen demand (BOD)
CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008
COD - Chemical oxygen demand (COD)
DMEL - Derived Minimal Effect level
DNEL - Derived-No Effect Level
EC50 - Median effective concentration
EC-No. - European Community number
EN - European Standard
IARC - International Agency for Research on Cancer
IATA - International Air Transport Association
IMDG - International Maritime Dangerous Goods
IOELV - Indicative Occupational Exposure Limit Value
LC50 - Median lethal concentration
LD50 - Median lethal dose
LOAEL - Lowest Observed Adverse Effect Level
N.O.S. - Not Otherwise Specified
NOAEC - No-Observed Adverse Effect Concentration
NOAEL - No-Observed Adverse Effect Level
NOEC - No-Observed Effect Concentration
OECD - Organisation for Economic Co-operation and Development
OEL - Occupational Exposure Limit
PBT - Persistent Bioaccumulative Toxic
PNEC - Predicted No-Effect Concentration
REACH - Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation (EC) No 1907/2006
RID - Regulations concerning the International Carriage of Dangerous Goods by Rail
SDS - Safety Data Sheet
STP - Sewage treatment plant
ThOD - Theoretical oxygen demand (ThOD)
TLM - Median Tolerance Limit
TRGS - Technical Rules for Hazardous Substances
VOC - Volatile Organic Compounds
vPvB - Very Persistent and Very Bioaccumulative
WGK - Water Hazard Class

Training advice : Normal use of this product shall imply use in accordance with the instructions on the packaging.

Other information : DISCLAIMER OF LIABILITY The information in this SDS was obtained from sources which we believe are reliable. However, the information is provided without any warranty, express or implied, regarding its correctness. The conditions or methods of handling, storage, use or disposal of the product are beyond our control and may be beyond our knowledge. For this and other reasons, we do not assume responsibility and expressly disclaim liability for loss, damage or expense arising out of or in any way connected with the handling, storage, use or disposal of the product. This SDS was prepared and is to be used only for this product. If the product is used as a component in another product, this SDS information may not be applicable.

Full text of H-statements & Precautionary statements (GHS UN):	
H303	May be harmful if swallowed
H313	May be harmful in contact with skin
H361d	Suspected of damaging the unborn child

H402	Harmful to aquatic life
P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.
P273	Avoid release to the environment.
P280	Wear protective gloves/protective clothing/eye protection/face protection/hearing protection/....
P308+P313	IF exposed or concerned: Get medical advice/attention.
P312	Call a POISON CENTER/doctor/physician if you feel unwell.
P405	Store locked up.
P501	Dispose of contents/container to in accordance with local regulations.

SDS UN - ETİ Maden

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

Safety Data Sheet author's

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